Which structures survive a two-dimensional view?

Data from Horowitz et al. (2013)

Mass cytometry (CyTOF): simultaneous protein-marker measurements on individual cells.

Provided donors	20
Analysis markers	37
Available gated NK cells	261,593
Balanced analysis sample	40,000

Our task

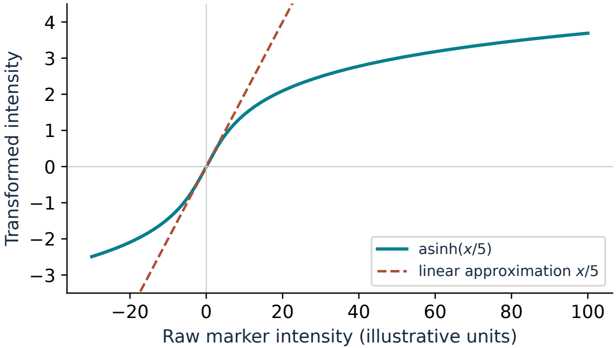
Visualise the same cells with **PCA, t-SNE and UMAP**; show parameter effects and compare the maps quantitatively.

Project context: CellCNN later tests whether cellular patterns predict donor CMV status. That performance comparison is not established by these maps.

CMV labels describe donors; they do not identify infected individual cells.

Horowitz et al., Sci. Transl. Med. 2013; Arvaniti & Claassen, Nat. Commun. 2017. Provided 20-donor subset; notebook 02_dimensionality_reduction_clean.

Transform intensities, then equalise marker scales



$$a_{ij} = \operatorname{asinh}(x_{ij}/5)$$

Approximately linear near zero; compresses large magnitudes and accepts negative background values.

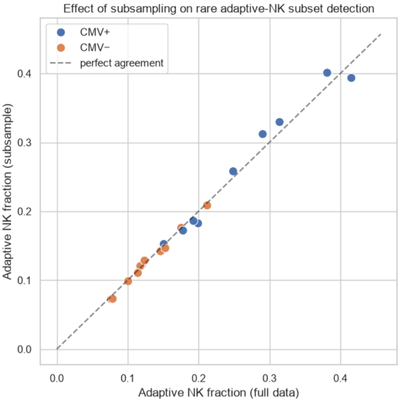
$$z_{ij} = \frac{a_{ij} - \bar{a}_j}{s_j}$$

Marker-wise z-scoring prevents large marker scales from dominating distances. All 37 markers pass the variance filter.

The geometry depends on preprocessing as well as on the projection method.

Notebook: cofactor 5; variance threshold 0.05 on the arcsinh scale; pooled scaling across the balanced exploratory sample. Curve is a mathematical illustration.

Does the balanced sample preserve the selected phenotype?



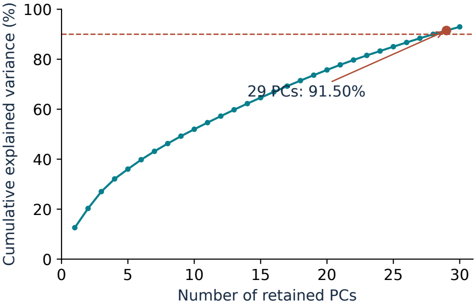
Compare **2,000 cells per donor** with all available gated NK cells from that donor.

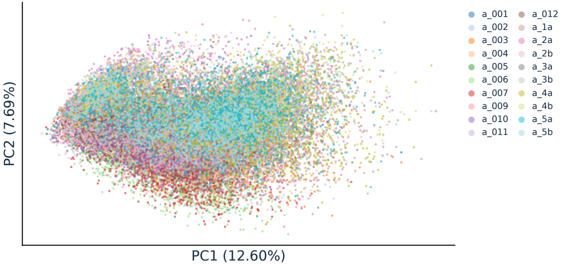
NKG2C$^+$/CD57$^+$ is a marker phenotype associated with **memory-like ("adaptive") NK cells** after CMV exposure. It is not itself a functional memory test.

Points lie close to the identity line. No donor has fewer than 10 flagged cells in the balanced sample.

Saved plot, cell 17 (counting from 1). Positivity: two-component GMM crossover per marker on all 261,593 cells. This check concerns one approximate phenotype, not every rare population.

29 PCs retain 91.50%; the plotted two retain only 20.29%



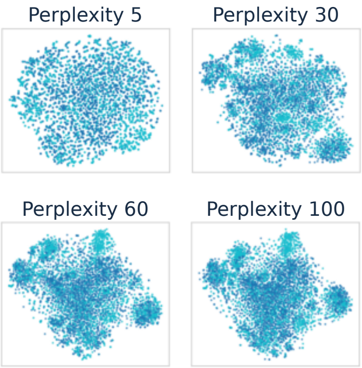


Retention choice: 29 PCs reach the 90% threshold and become the input to t-SNE/UMAP.

Display choice: PC1/PC2 show substantial donor overlap, but omit most retained variance.

40,000 cells, arcsinh + z-scoring. Spectrum verified from the same 37-marker covariance matrix; donor plot uses committed PCA coordinates. Notebook counterparts: cells 20–22.

t-SNE: changing perplexity changes the neighbourhood scale



5,000 fixed cells, 29 PCs: reduce sweep time and memory.

Perplexity is an effective neighbourhood size, not an exact neighbour count.

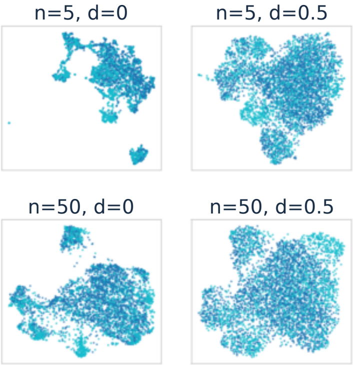
Selected: 60, using the largest saved Trustworthiness:

50	0.925973
60	**0.926046**
100	0.925800

T: penalises false neighbours by their original ranks.
R: fraction of original neighbours retained.
The choice here uses T alone.

Selected panels from saved cell 29; colours indicate donor CMV status. The very small score gap is not evidence of a stable optimum. Invalid saved sweep kNN scores are omitted.

UMAP: neighbourhood scale versus point packing



n_neighbors: size of the local neighbourhood used to construct the graph.

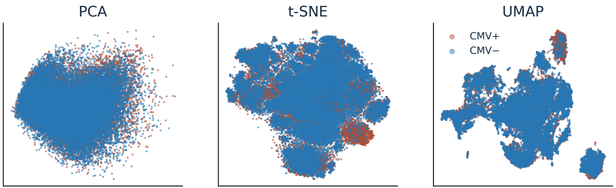
min_dist: controls how tightly points can pack in the embedding.

$n_{\text{neighbors}}$	min_dist	$T(15)$
5	0.0	0.890139
5	0.1	0.879145
30	0.0	0.877151

Selected: **5 neighbours, min_dist = 0.0**. Compact islands do not prove distinct cell types.

Four corner settings from the 16-panel saved grid, cell 33; colours indicate CMV status. Top three rows ranked by saved Trustworthiness on the same 5,000 cells and 29 PCs.

Local preservation and global distances favour different maps



Embedding	$T(15)\uparrow$	$R(15)\uparrow$	$r_d\uparrow$	$S_{\mathrm{CMV}}\uparrow$
PCA	0.7258	0.0072	**0.6204**	0.0016
t-SNE	**0.9612**	**0.2232**	0.4868	0.0104
UMAP	0.9009	0.1126	0.3954	**0.0178**

T, R: local structure. r_d: Pearson correlation of pairwise distances. S_{CMV}: separation by donor CMV label.

t-SNE leads locally; PCA leads in distance correlation. All CMV silhouettes are close to zero.

40,000 cells; reference: 29 PCs; k=15. Distance correlation: fixed 2,000-cell sample. T and silhouette are saved outputs; R is corrected.
Green/red denote numerical best/worst within a column, not clinical validity.

Which maps are most similar? It depends on the measure

Pair	$R(15)\uparrow$	$M^2\downarrow$
PCA vs t-SNE	0.00585	**0.45189**
PCA vs UMAP	0.00410	0.73390
t-SNE vs UMAP	**0.12532**	0.72369

Chosen task measure: Procrustes disparity M^2.

Align two maps by translation, rotation/reflection and scaling; measure the remaining mismatch.

Local neighbours: t-SNE and UMAP are the most similar pair.

Aligned global shape: PCA and t-SNE are the most similar pair.

These answers differ because the measures ask different questions.

Use maps to generate hypotheses; assess clusters and donor-level predictions in the later tasks.

All three pairs evaluated on the same 40,000 cells in the same order. R(15) corrected from saved coordinates; SciPy Procrustes values independently reproduced. Neither metric identifies biologically correct cell types.

Backup: how t-SNE turns neighbours into a map

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2/(2\sigma_i^2))}{\sum_{k\neq i}\exp(-\|x_i - x_k\|^2/(2\sigma_i^2))}, \quad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n},$$

$$q_{ij} = \frac{(1+\|y_i - y_j\|^2)^{-1}}{\sum_{k\neq l}(1+\|y_k - y_l\|^2)^{-1}}, \qquad C = \sum_{i\neq j} p_{ij}\log\frac{p_{ij}}{q_{ij}}.$$

Bandwidth σ_i matches the target perplexity $2^{H(P_i)}$. The heavy-tailed kernel helps alleviate crowding; optimisation minimises KL divergence.

Varied	Perplexity: 5, 30, 50, 60, 70, 100.
Fixed	2 dimensions; PCA initialisation; seed 42; 750 iterations; Barnes–Hut method.
Defaults	Euclidean metric, early exaggeration 12, automatic learning rate, angle 0.5.

Other parameters were held fixed to limit the experiment; they can also change the map.

van der Maaten & Hinton (2008); scikit-learn TSNE documentation. Fixed iteration count alone is not a convergence check. Defaults refer to the documented scikit-learn 1.9 environment.

Backup: two local metrics, two different penalties

Let $N_i^X(k)$ and $N_i^Y(k)$ be the k neighbours of cell i, excluding itself.

$$R(k) = \frac{1}{n} \sum_{i=1}^{n} \frac{|N_i^X(k) \cap N_i^Y(k)|}{k}$$

Symmetric; exact neighbour identities matter. Ten shared neighbours out of fifteen contribute 10/15. This is overlap/recall, not Jaccard (10/20 in that example).

$$T(k) = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i} \sum_{j \in N_i^Y(k) \setminus N_i^X(k)} (r_X(i,j) - k).$$

Penalises new neighbours according to their rank in the reference. A point ranked 16th incurs a smaller penalty than one ranked 1,000th; direction matters.

Correction: requesting neighbours with no explicit query already excludes self. Removing the first returned column retained 14 neighbours but divided by 15. The corrected identity score is 1.

Venna & Kaski (2001); scikit-learn trustworthiness and NearestNeighbors documentation. Here k=15. Full-sample scores average cells; equal cells per donor give equal donor weight.

Backup: how UMAP constructs and embeds a weighted graph

$$w_{j|i} = \exp\left[-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right], \qquad w_{ij} = w_{j|i} + w_{i|j} - w_{j|i}w_{i|j}.$$

Offset ρ_i and scale σ_i adapt to local density; directed neighbourhoods are combined into symmetric weights.

$$v_{ij} = (1 + a\|y_i - y_j\|^{2b})^{-1}, \qquad L \sim -\sum_{i<j}\{w_{ij}\log v_{ij} + (1 - w_{ij})\log(1 - v_{ij})\}.$$

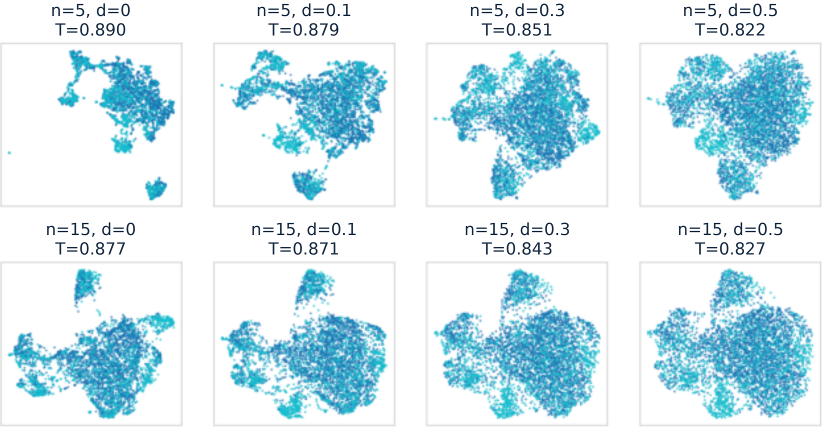
Sampled updates balance attraction and repulsion. The parameters a, b depend on min_dist and spread.

Swept	n_neighbors: 5, 15, 30, 50; min_dist: 0, 0.1, 0.3, 0.5.
Fixed/default	2 dimensions, seed 42, Euclidean metric, spectral initialisation, spread 1.

Larger neighbourhoods change the scale of the graph; larger min_dist discourages tight packing. Neither guarantees accurate global distances.

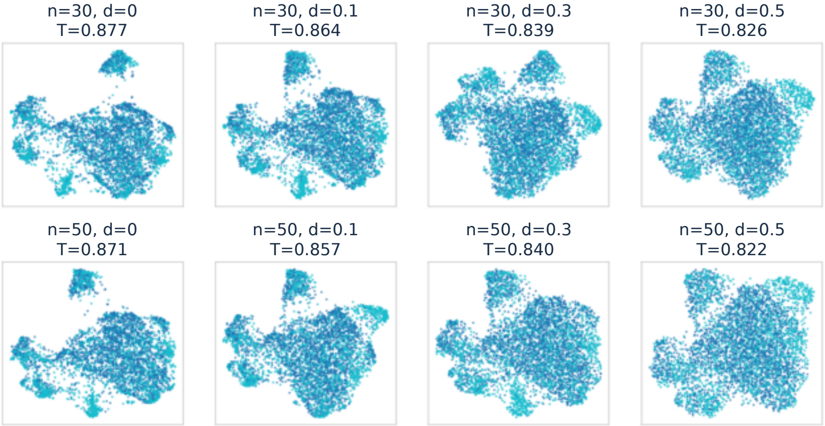
McInnes, Healy & Melville (2018/2020); official UMAP parameter and algorithm documentation. Equations describe the conceptual objective, not every optimisation approximation.

Backup: UMAP sweep, 5 and 15 neighbours



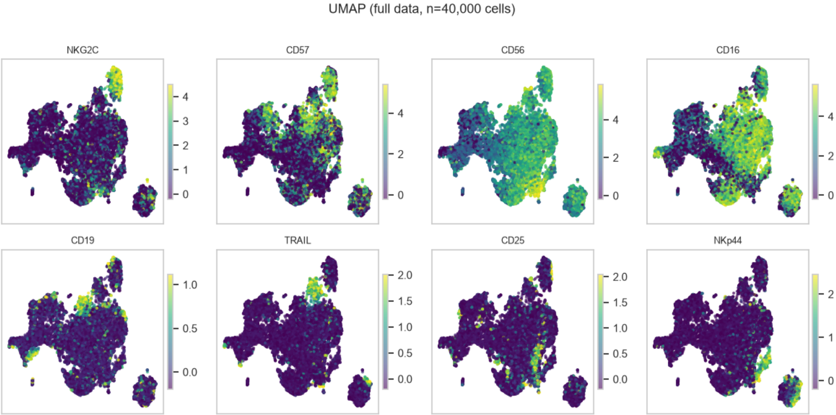
First two rows of the original 16-panel grid. Columns: min_dist = 0, 0.1, 0.3, 0.5. Same 5,000 cells, 29 PCs and seed 42; each panel reports saved Trustworthiness.

Backup: UMAP sweep, 30 and 50 neighbours



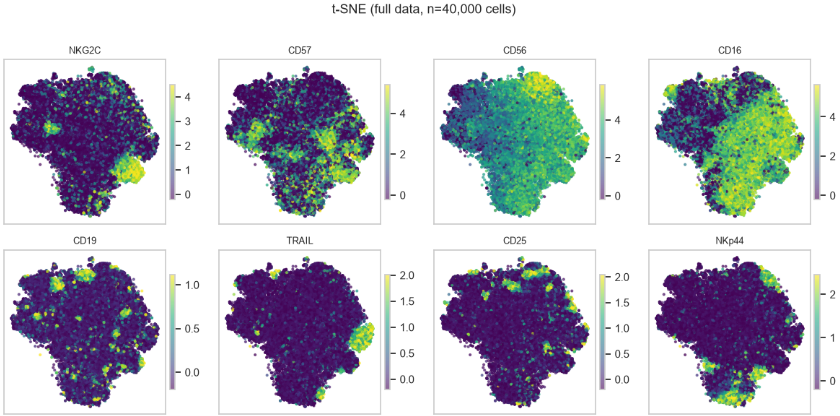
Last two rows of the original 16-panel grid. Columns: min_dist = 0, 0.1, 0.3, 0.5. Visual changes are parameter effects within one run, not a stability study across seeds.

Backup: marker expression on the UMAP map



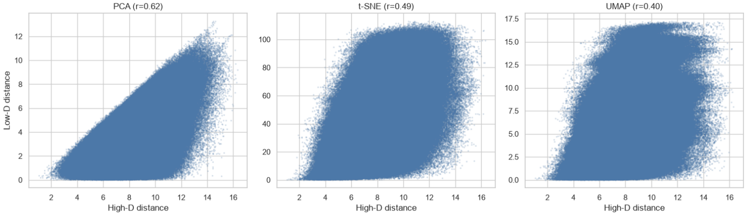
Original saved output, cell 41: 40,000 balanced cells, not all 261,593 available NK cells. Arcsinh-transformed marker values; colour limits use the 1st/99th percentiles separately for each marker.

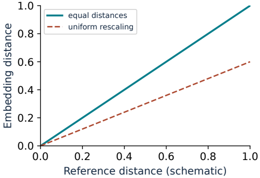
Backup: the same markers on the t-SNE map



Original saved output, cell 41. Colour patterns are exploratory expression gradients. A bright region does not establish an infected-cell population or a validated adaptive-NK gate.

Backup: Shepard diagrams compare distances, not neighbours





Each point is a pair of cells. $r_d = \mathrm{corr}(d_X, d_Y)$ uses all pairs among 2,000 sampled cells.

Exact distance equality gives the identity line. Uniform rescaling gives another straight line with $r_d = 1$; therefore the diagonal is not required for perfect correlation.

Top: saved notebook diagrams, 1,999,000 distances per representation. Bottom: schematic illustration, not experimental data. Pearson distance correlation is not the distance-correlation statistic of independence testing.

Backup: Procrustes aligns the maps before measuring mismatch

Centre and normalise each coordinate matrix to unit Frobenius norm:

$$A_c = \frac{A - \bar{A}}{\|A - \bar{A}\|_F}, \qquad B_c = \frac{B - \bar{B}}{\|B - \bar{B}\|_F}.$$

$$M^2 = \min_{s, Q:\, Q^\top Q = I} \|A_c - sB_cQ\|_F^2.$$

- Rows must correspond to exactly the same cells in the same order.
- Rotation/reflection, translation and uniform scaling do not count as biological differences.
- Lower disparity means a closer global fit after alignment; it does not establish local fidelity or class separation.
- Nonlinear deformations can change neighbour identities and global shape differently.

SciPy scipy.spatial.procrustes. With its normalisation and optimal scaling, the scalar disparity is symmetric even though the returned transformed matrices have different normalisation conventions.

Backup: sources and verification scope

Data and biology

Horowitz et al. (2013). *Genetic and environmental determinants of human NK cell diversity revealed by mass cytometry.* Sci. Transl. Med. 5:208ra145.

Arvaniti & Claassen (2017). *Sensitive detection of rare disease-associated cell subsets via representation learning.* Nat. Commun. 8:14825.

Methods

van der Maaten & Hinton (2008), JMLR 9:2579–2605 (t-SNE).

McInnes, Healy & Melville (2018/2020), arXiv:1802.03426 (UMAP).

Venna & Kaski (2001), ICANN, pp. 485–491 (Trustworthiness).

Official scikit-learn and SciPy documentation; linked references in the detailed report.

Provenance

Notebook source: main commit 252415a. kNN corrected from committed coordinates; Procrustes and distance correlations checked. Embeddings, sweep Trustworthiness and CMV silhouettes were not retrained/recomputed. No new classification benchmark was run.